

An Improved Algorithm for Sorting by Block-Interchanges Based on Permutation Groups

Yen-Lin Huang¹, Cheng-Chen Huang², Chuan-Yi Tang¹ and
Chin-Lung Lu²

¹Department of Computer Science, National Tsing Hua
University, Hsinchu 300, Taiwan

²Institute of Bioinformatics Department of Biological Science
and Technology, Nation Chiao Tung University, Hsinchu 300,
Taiwan

The 26 th Workshop on Combinatorial Mathematics and
Computation Theory, PP. 9-14, Chiayi, Taiwan, Apr. 24-25, 2009

Presenter: Hung-Hsi Chen
Date: Jun. 02, 2009

Abstract

Given a chromosome represented by a permutation of genes, a **block-interchanges** is proposed to as a generalized transposition that **affects a chromosome by swapping two nonintersecting segments of genes**. The problem of sorting by block-interchanges is to find a minimum series of block-interchanges for transforming one chromosome into another. In this paper, we present an $O(n+\delta \log n)$ time algorithm for solving the problem of sorting by block-interchanges, which improves a previous algorithm of $O(\delta n)$ time proposed by Lin et al.[12], where n is the number of genes and δ is the minimum number of block interchanges required for the transformation.

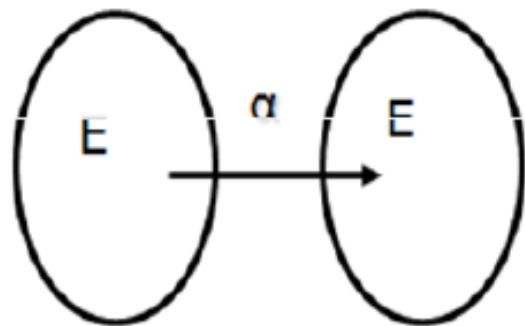
Introduction

- **Genome rearrangement** : to measure the evolutionary difference between two organisms
- Unlike from the traditional point mutations such as insertions, deletions, and substitutions, various **large-scale mutations**, such as **reversals**, **transpositions**, **block-interchanges**, **fissions** and **fusions** and **translocations** have been proposed to **determine the evolutionary distance** between two related genomes by comparing the **gene orders**.

- Christie (1996) : first introduced the concept of block-interchange and also proposed an $O(n^2)$ time algorithm using the breakpoint graph approach, where n is the number of genes.
- Lin et al. (2005): use of permutation groups in algebra, designed a algorithm for solving the same problem on both linear or circular chromosomes with time-complexity of $O(\delta n)$, where δ is the minimum number of block-interchanges required for the transformation and can be calculated in $O(n)$ time in advance.
- In this paper, we present an $O(n + \delta \log n)$ time algorithm for solving the problem of sorting by block-interchanges, where δ is the same as above.

Permutation group in algebra

A **permutation** α is a 1-1 mapping from a set E into itself.



$E = \{1, 2, 3, 4, 5\}$

$$\alpha(1) = 3$$

$$\alpha(2) = 5$$

$$\alpha(3) = 2$$

$$\alpha(4) = 1$$

$$\alpha(5) = 4$$



$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 2 & 1 & 4 \end{bmatrix}$$

Array form



$$(1, 3, 2, 5, 4)$$

Cycle form

$$= (3, 2, 5, 4, 1)$$

$$= (2, 5, 4, 1, 3)$$

$$= (5, 4, 1, 3, 2)$$

$$= (4, 1, 3, 2, 5)$$

5-cycle

Permutation(cycle) property

- **Disjoint** : two cycles have no element in common.

Ex : $C1 = (1, 2, 3, 4, 5)$, $C2 = (6, 7)$ are disjoint.

- **K-cycle** : cycle of length K

Ex : $(3, 4, 5, 6)$ is 4-cycle, $(1, 2)$ is 2-cycle

- The cycle of **length one** is not explicitly written and its element is said to be **fixed**.

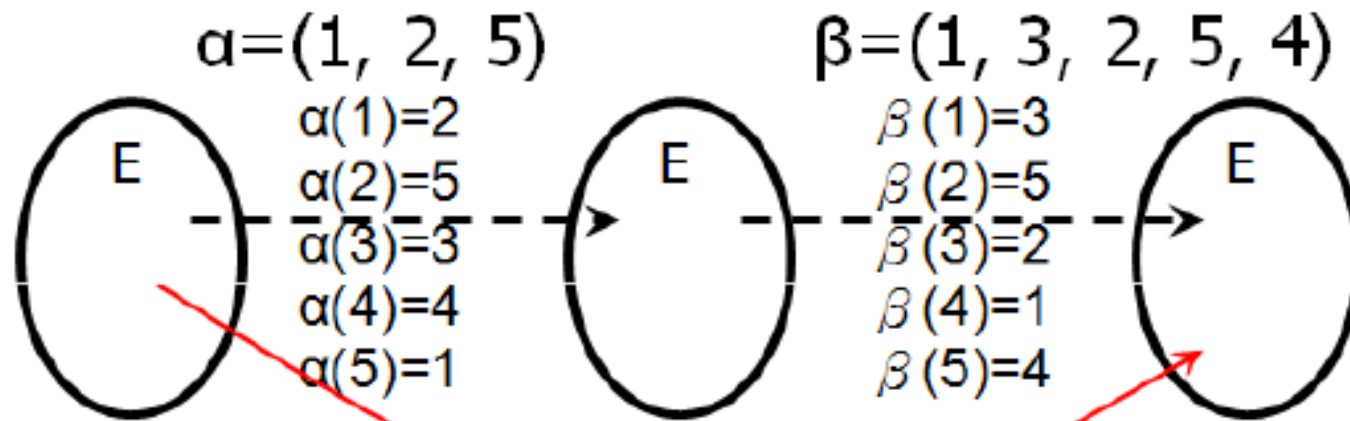
Ex : $(1), (2), (3)$

- Any permutation can be written in a **unique way as a product of disjoint cycles**, if the order of cycles is ignored.

Ex :

$$\left[\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{array} \right] = (4,5)(1)(2,3) \\ = (1)(2,3)(4,5) = (2,3)(4,5) \\ = (2,3)(4,5)(1)$$

Composition(product) of permutations



$$\beta \alpha = \beta(\alpha(x))$$

$$= (1, 3, 2, 5, 4) (1, 2, 5)$$

$$= (5, 3, 2, 4, 1)$$

Note: every permutation can be written as a product of 2-cycle

$$\text{Ex : } (1, 2, 3, 4) = (1, 4)(1, 3)(1, 2)$$

Inverse of a permutation

$$\alpha\alpha^{-1} = \alpha^{-1}\alpha = \mathbf{1} = (1) (2) \dots (n)$$

$$\alpha = (1, 4, 2, 3) (5, 6)$$



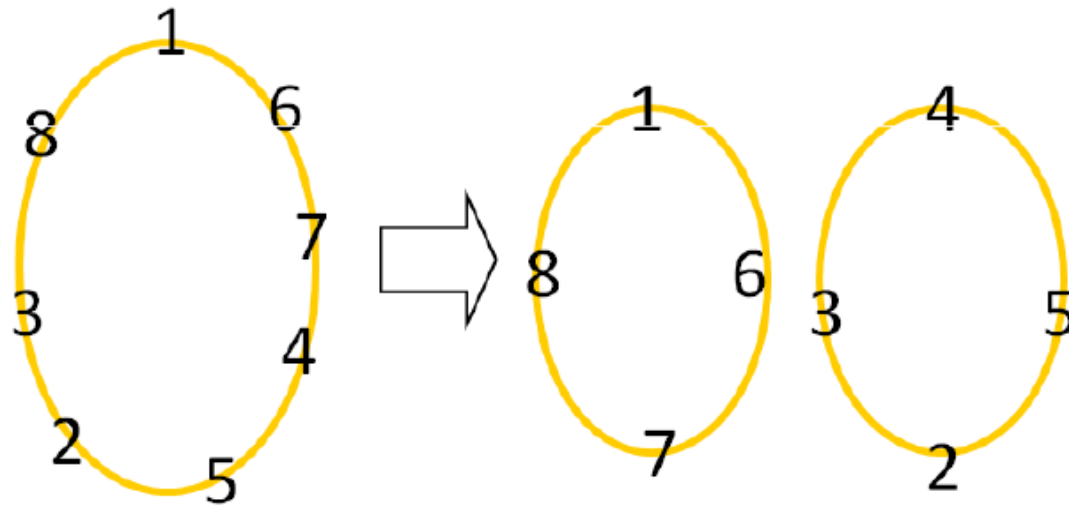
$$\alpha^{-1} = (3, 2, 4, 1) (6, 5)$$

Note : if $C = (x, y)$ is 2-cycle, $C^{-1} = (x, y)$

Fission/split into two cycles

If $p = (x, y)$ is 2-cycle and x, y are in the same cycle of permutation α then $p\alpha$ is broken into two disjoint cycles.

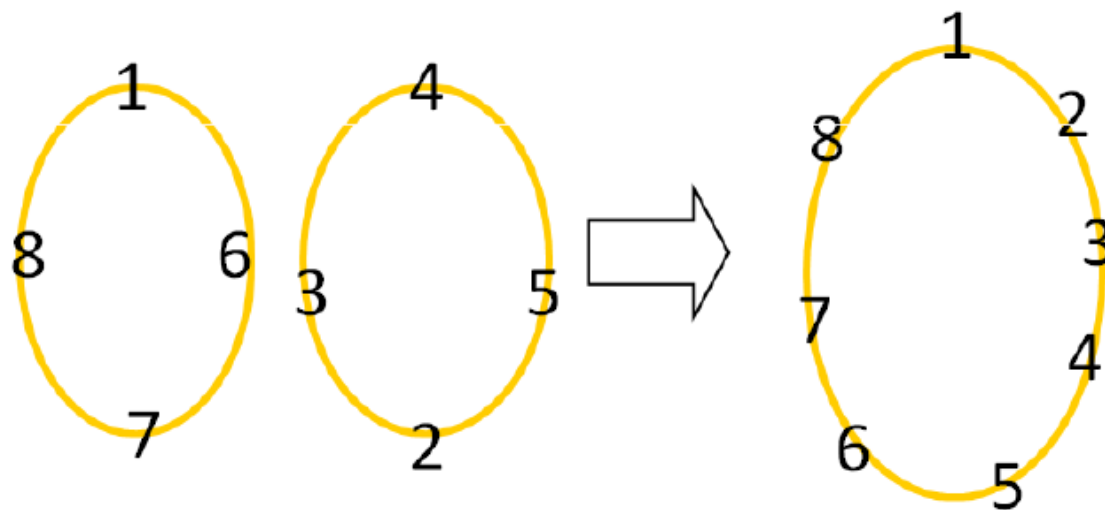
$$(4, 8) (\underline{4}, 5, 2, 3, \underline{8}, 1, 6, 7) = (4, 5, 2, 3)(8, 1, 6, 7)$$



Fusion/join two cycles together

If $p = (x,y)$ is 2-cycle and x, y are in different cycle of permutation α then $p\alpha$ join two disjoint cycles together.

$$(2,6) (\underline{2},3,4,5) (\underline{6},7,8,1) = (2,3,4,5,6,7,8,1)$$



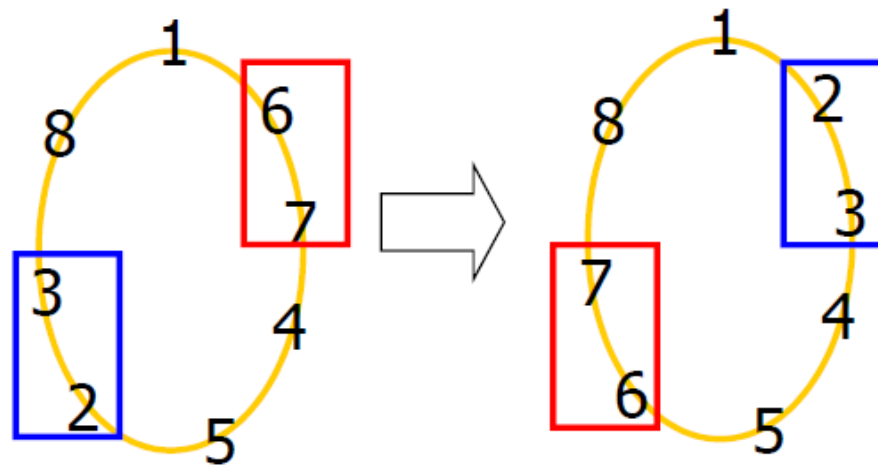
Block interchange via two 2-cycles

Join Split

$$(2,6)(4,8)(4,5,2,3,8,1,6,7) = (2,6)(4,5,2,3)(8,1,6,7)$$

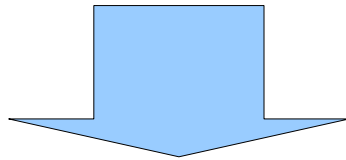
Join Split

$$(2,6)(4,8)(4,5,2,3,8,1,6,7) = (2,3,4,5,6,7,8,1)$$

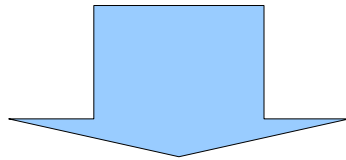


Information of block-interchanges

$$P_k P_{k-1} \dots P_1 (1, 6, 7, 4, 5, 2, 3, 8) = (1, 2, 3, 4, 5, 6, 7, 8)$$



$$(P_k P_{k-1} \dots P_1) \cdot \alpha = I$$

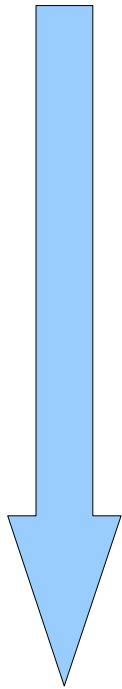


$$\begin{aligned} (P_k P_{k-1} \dots P_1) &= I \alpha^{-1} \\ &= (1, 2, 3, 4, 5, 6, 7, 8)(8, 3, 2, 5, 4, 7, 6, 1) \\ &= (2, 6)(4, 8) \end{aligned}$$

Lin's algorithm

$$\alpha = (7, 1, 4, 6, 2, 5, 3, 8)$$

$$I\alpha^{-1} = (1, 8, 4, 2, 7)(3, 6, 5)$$



Step 1 : $O(n)$

The **block-change distance** between α and I

$$\delta = (n - f(I\alpha^{-1})) / 2 = 3$$

Where $f(x)$ denotes **the number of disjoint cycles** in the cycle decomposition of x .

$$I = (1, 2, 3, 4, 5, 6, 7, 8)$$

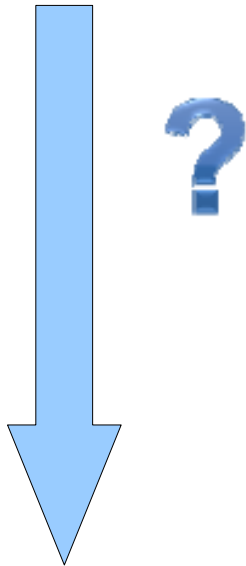
Step 2 : $O(\delta n)$

Find the three block-interchange operations

$$\sigma_1 = p_{12}p_{11}, \sigma_2 = p_{22}p_{21}, \sigma_3 = p_{32}p_{31}$$

$$\begin{aligned}\alpha &= (7, 1, 4, 6, 2, 5, 3, 8) \\ &= (1, 4, 6, 2, 5, 3, 8, 7)\end{aligned}$$

$$I\alpha^{-1} = (\underline{1, 8}, 4, 2, 7)(3, 6, 5)$$



Let $p_{11} = (1, 8)$ (split operation)
since 1 and 8 are adjacent in $I\alpha^{-1}$

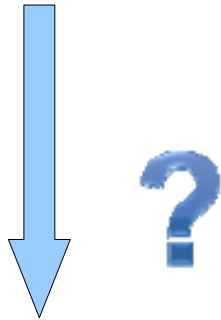
$$I = (1, 2, 3, 4, 5, 6, 7, 8)$$

$$\alpha = (1, 4, 6, 2, 5, 3, 8, 7)$$

$$I\alpha^{-1} = (1, 8, 4, 2, 7)(3, 6, 5)$$

$$\begin{aligned} p_{11} \alpha &= (1, 8)(1, 4, 6, 2, 5, 3, 8, 7) \\ &= (1, 4, 6, 3, 5, 3)(8, 7) \end{aligned}$$

$$\begin{aligned} I\alpha^{-1}p_{11}^{-1} &= I\alpha^{-1}p_{11} \\ &= (1, 8, 4, 2, 7)(3, 6, 5)(1, 8) \\ &= (1, 4, \underline{2, 7})(3, 6, 5) \end{aligned}$$



$$I = (1, 2, 3, 4, 5, 6, 7, 8)$$

$\therefore \text{index}(8) = 7$ in α (1 as first element in α) $\therefore k = 7$

Find $P_{12} = (u, v)$ where u, v are adjacent elements in $I\alpha^{-1}p_{11}$ and $\text{index}(u) \leq k-1, \text{index}(v) \geq k$

$$P_{12} = (2, 7) \text{ (join operation)}$$

$$\alpha = (1, 4, 6, \boxed{2, 5, 3}, 8, \boxed{7})$$

$$P_{12}p_{11}\alpha = \boxed{2, 5, 3}, 1, 4, 6, \boxed{7}, 8)$$



$$I = (1, 2, 3, 4, 5, 6, 7, 8)$$

$$I\alpha^{-1} = (1, 8, 4, 2, 7)(3, 6, 5)$$

$$I\alpha^{-1}p_{11}p_{12} = (1, 4, 2)(3, 6, 5)$$

In the **second iteration**, we have $P_{21} = (1, 4)$ (split operation)

$$P_{21}P_{12}p_{11}\alpha = (1, 4)(2, 5, 3, 1, 4, 6, 7, 8) = (1)(4, 6, 7, 8, 2, 5, 3)$$

$$I\alpha^{-1}p_{11}p_{12}P_{21} = (1, 4, 2)(3, 6, 5)(1, 4) = (1, 2)(3, 6, 5)$$

Find $P_{22} = (1, 2)$ (join operation)

$$P_{22}P_{21}P_{12}p_{11}\alpha = (4, 6, 7, 8, 1, 2, 5, 3)$$

$$I\alpha^{-1}p_{11}p_{12}P_{21}P_{22} = (3, 6, 5)$$

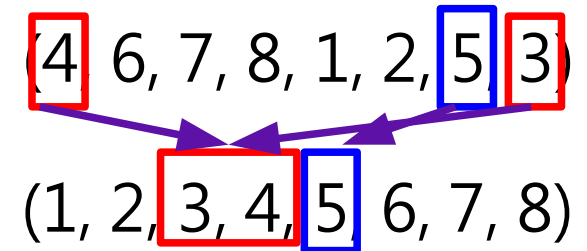
$$P_{12}p_{11}\alpha = \boxed{2, 5, 3}, \boxed{1}, 4, 6, 7, 8)$$

$$P_{22}P_{21}P_{12}p_{11}\alpha = (4, 6, 7, 8, \boxed{1}, \boxed{2, 5, 3})$$

In the **third iteration**, we have $P_{31} = (3, 6)$ (split operation)

$$P_{31}P_{22}P_{21}P_{12}p_{11}\alpha = (4, 3)(1, 2, 5, 6, 7, 8)$$

$$I\alpha^{-1}p_{11}p_{12}P_{21}P_{22}P_{31} = (3, 5)$$



Find $P_{32} = (3, 5)$ (join operation)

$$P_{32}P_{31}P_{22}P_{21}P_{12}p_{11}\alpha = I = (1, 2, 3, 4, 5, 6, 7, 8)$$

$$I\alpha^{-1}p_{11}p_{12}P_{21}P_{22}P_{31}p_{32} = 1 = (1)(2)(3)(4)(5)(6)(7)(8)$$

After 3 iterations, we interchange α to $(1, 2, 3, 4, 5, 6, 7, 8)!!$

Time complexity: $O(n) + O(\delta n) = O(\delta n)$

Lemma

Let α' be a permutation of $E = \{1, 2, \dots, n\}$, C be a cycle of α' , and z be the maximum element in C . Suppose that $z \neq n$. Then $\alpha'(z)$ and $I(z)$ are adjacent in $I\alpha'^{-1}$, and $(\alpha'(z), I(z))$ is a join operation of α' .

pf :

- $\therefore I\alpha'^{-1}(\alpha'(z)) = I(z)$
- $\therefore \alpha'(z), I(z)$ are adjacent in $I\alpha'^{-1}$
- $\therefore z$ and $\alpha'(z)$ are in the same cycle of α'
- $\therefore \alpha'(z), I(z)$ are in different cycle of α'
- $\therefore (\alpha'(z), I(z))$ is a join operation of α'
- $\therefore z$ is maximum in C and $I(z) = z+1$
- $\therefore I(z)$ is not in C
- $\therefore z$ and $I(z)$ are in different cycle of α'

Ex :

$$\alpha' = (1, 4, 6, 2, 5, 3)(8, 7)$$

$$I\alpha'^{-1} = (1, 4, 2, 7)(3, 6, 5)$$

Find $z = 6$

$$\alpha'(z) = 2, I(z) = 7$$

are adjacent in $I\alpha'^{-1}$

$(2, 7)$ is join operation of α'

Permutation tree

Permutation tree of $\pi = (\pi_1, \pi_2, \dots, \pi_n)$

definition :

- 1 、 balanced tree with n leaves
- 2 、 the leaves are labelled by $(\pi_1, \pi_2, \dots, \pi_n)$, respectively.
- 3 、 internal nodes are labelled in a way such that labels of parent nodes are greater than those of their children.

$\therefore$ root has the maximum element in π

Three operations applied to the permutation tree :

Build : build a permutation tree, $O(n)$

Join : join two permutation trees into one, $O(\log n)$

Split : split one permutation tree into two, $O(\log n)$

New algorithm

$$\alpha = (7, 1, 4, 6, 2, 5, 3, 8)$$

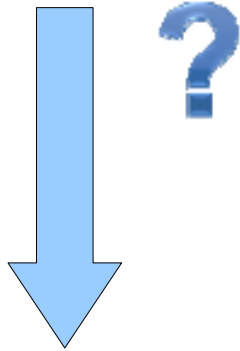
$$I\alpha^{-1} = (1, 8, 4, 2, 7)(3, 6, 5)$$

$$\delta = (n - f(I\alpha^{-1})) / 2 = 3$$

$$p_{11} = (1, 8) \text{ (split operation)}$$

$$p_{11}\alpha = (1, 8)(7, 4, 6, 2, 5, 3, 8) \\ = (1, 4, 6, 2, 5, 3)(8, 7)$$

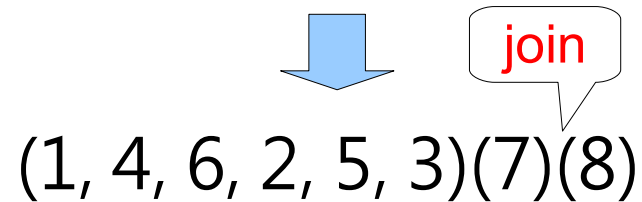
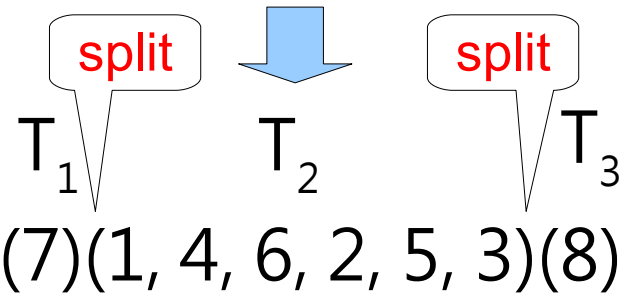
fission



$$I = (1, 2, 3, 4, 5, 6, 7, 8)$$

Permutation tree :

$$(7, 1, 4, 6, 2, 5, 3, 8)$$



$$(1, 4, 6, 2, 5, 3)(8, 7)$$

$$P_{11}\alpha = (1, 4, 6, 2, 5, 3)(8, 7)$$

$$I\alpha^{-1}P_{11} = (1, 4, 2)(3, 6, 5)$$

From lemma, we get $z = 6$

$$P_{12} = (\alpha(z), I(z)) = (2, 7)(\text{join operation})$$

$$P_{12}P_{11}\alpha = (2, 7)(1, 4, 6, 2, 5, 3)(8, 7) = (2, 5, 3, 1, 4, 6, 7, 8)$$

fusion

Permutation tree :

$$(1, 4, 6, 2, 5, 3)(8, 7)$$

$$\begin{array}{c} \text{split} \quad \downarrow \quad \text{split} \\ (1, 4, 6)(2, 5, 3)(8)(7) \end{array}$$

$$\begin{array}{c} \text{join} \quad \downarrow \quad \text{join} \quad \text{join} \\ (2, 5, 3)(1, 4, 6)(7)(8) \end{array}$$

$$\downarrow$$

$$(2, 5, 3, 1, 4, 6, 7, 8)$$

Time complexity

- Build permutation tree: $O(n)$
- Calculate δ : $O(n)$
- Loop δ times
 - Find split operation : $O(1)$
 - Find maximum element in cycle(to find join operation, using permutation tree) : $O(\log n)$
 - Compute split and join operation on α . : $O(\log n)$
 - Computer split and join operation on $I\alpha^{-1}$: $O(\log n)$
- End loop

Time complexity: $O(n + \delta \log n)$

Thank You!!